



Tyk Architect Certification n

certification preparation Course



Tyk Deployment Models

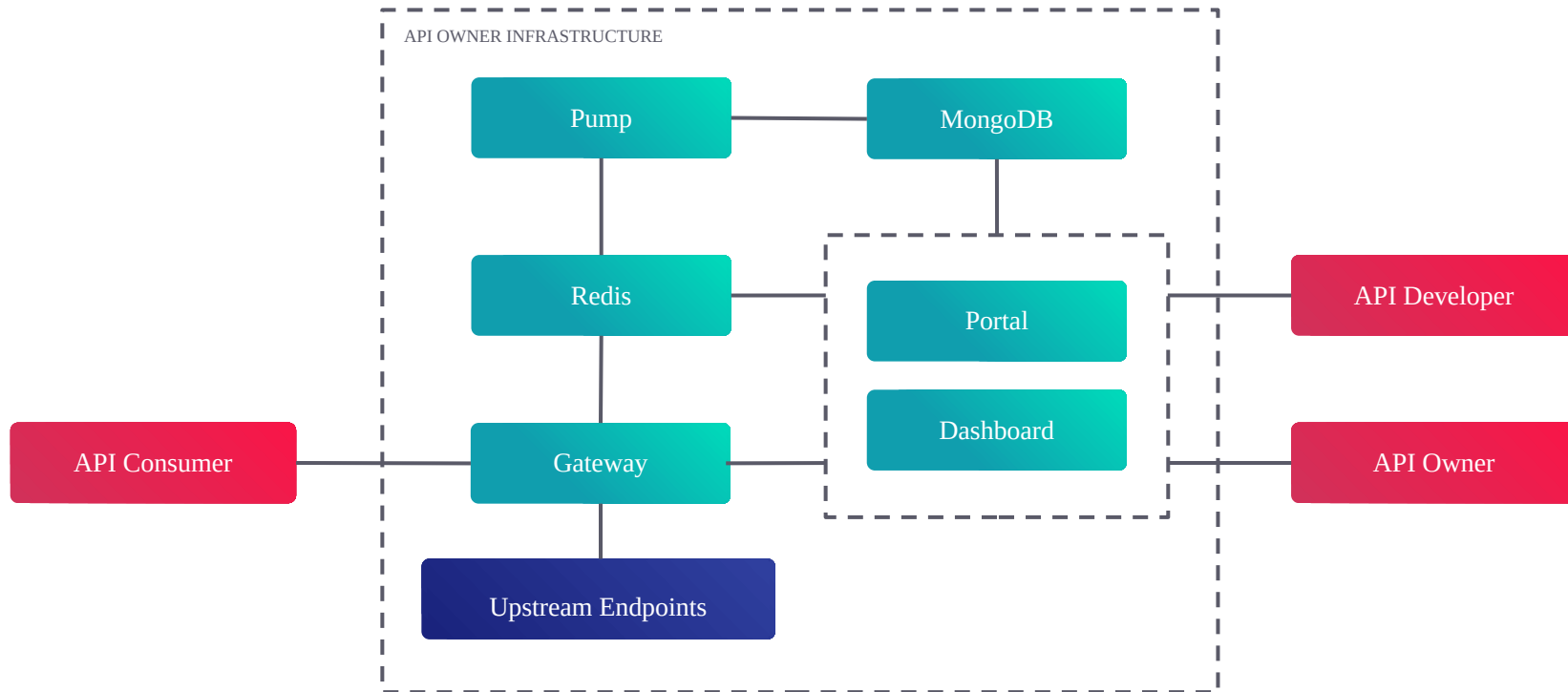
Tyk Deployment Models

- **On-premise**
 - Tyk solution hosted on customer's infrastructure
- **Cloud**
 - Tyk solution hosted on Tyk's infrastructure
- **Hybrid Cloud**
 - Tyk solution hosting split between Tyk and customer infrastructure
- **Hybrid On-Premise**
 - Tyk solution hosted on customer infrastructure spanning multiple data centers and/or regions

Tyk Deployment Models: On-Premise

- **Entire solution hosted on customer's infrastructure**
- **Full control over how Tyk is deployed and maintained**
- **Fully customizable developer portal**
- **Architecture limited to single data center**
- **Install Options**
 - Platforms: Docker, Kubernetes, OpenShift
 - OS Support: RHEL/Centos, Debian/Ubuntu, Amazon Linux
 - Cloud Host(s): Amazon AWS, GCP, Microsoft Azure, Heroku

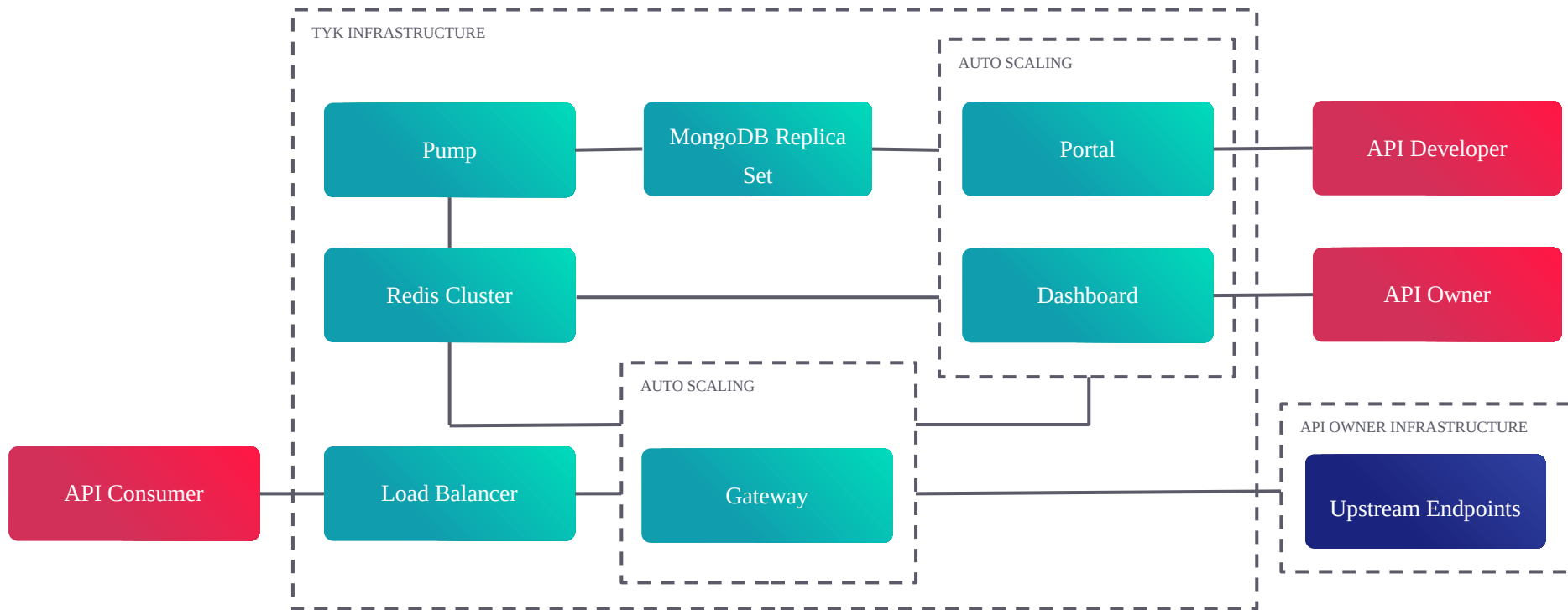
Tyk Deployment Models: On-Premise Architecture



Tyk Deployment Models: Cloud

- **Solution hosted within Tyk infrastructure**
- **SaaS product, no installation required**
- **Single tenant, cloud infrastructure not shared with other clients**
- **Customer can choose which AWS region(s) to host solution**
 - Regions: Singapore (ap-southeast-1), Frankfurt (eu-central-1), London (eu-west-2), N. Virginia (us-east-1), and Oregon (us-west-2)
 - Dashboard and Gateway(s) can be deployed to different regions

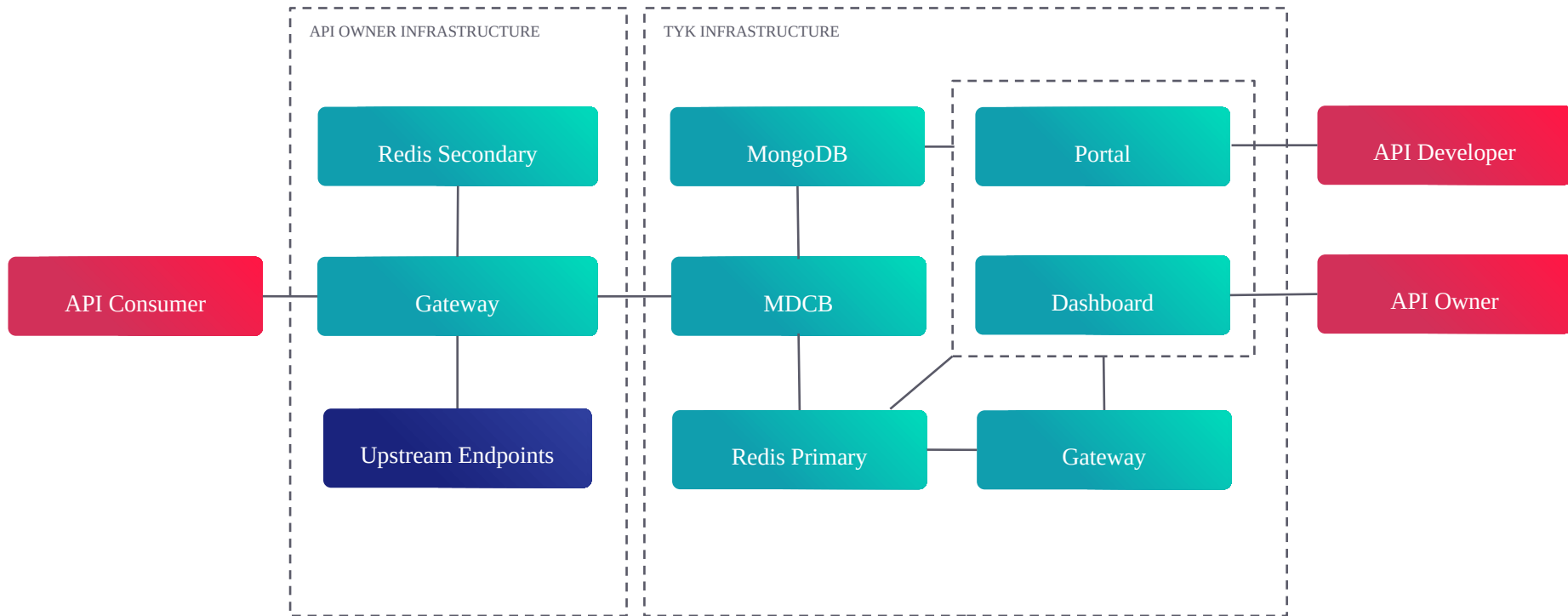
Tyk Deployment Models: Cloud Architecture



Tyk Deployment Models: Hybrid Cloud

- **Hosting split between Tyk and customer's infrastructure**
 - Tyk Hosts: Dashboard, Portal, MongoDB, and the primary Redis instance
 - Customer Hosts: Gateways and local Redis instance(s)
 - Allows customers to put their gateways in closer proximity to their upstream endpoints
 - Gateways and local Redis instance(s) is/are ephemeral so these can be spun up and down without worry about the loss of data, as all data is persisted on the primary Redis instance
 - All API traffic is routed via client's infrastructure, no API traffic passes through Tyk's network
- **Requires the Multi Data Centre Bridge component**

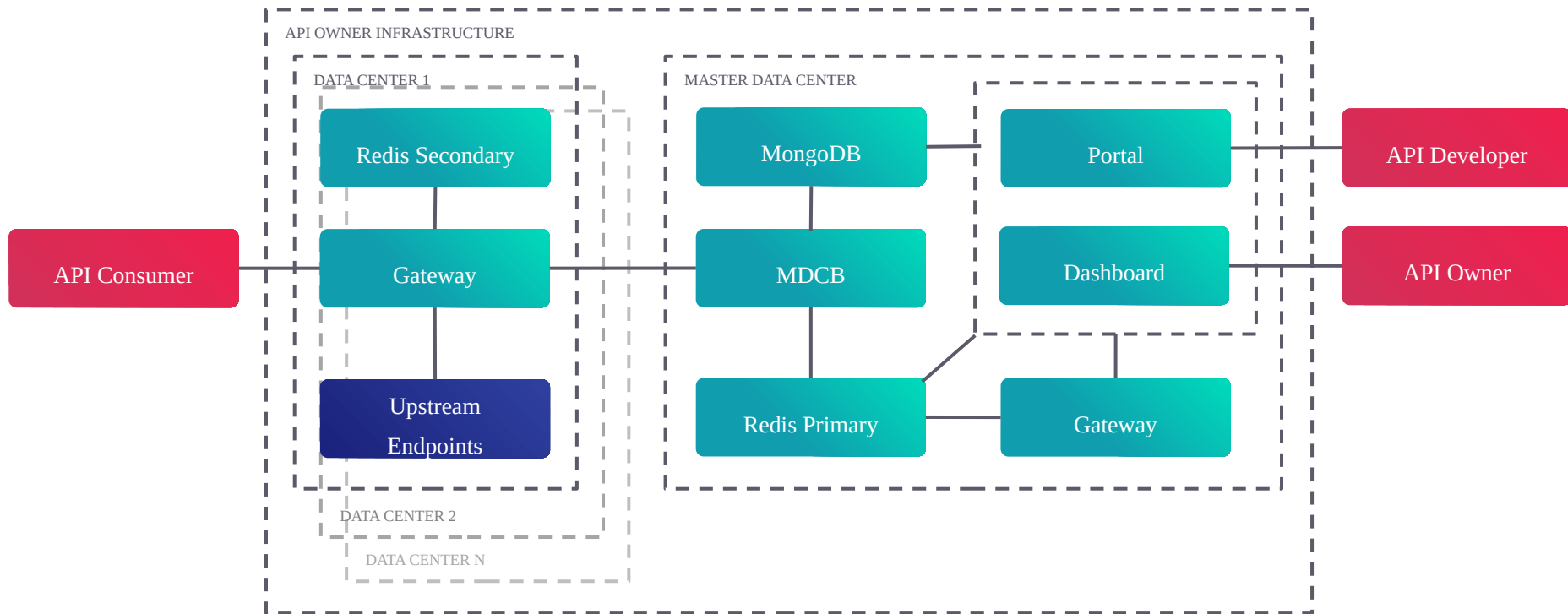
Tyk Deployment Models: Hybrid Cloud Architecture



Tyk Deployment Models: On-Premise

- **Solution hosted entirely on client's infrastructure with no reliance on Tyk Cloud**
- **Clients responsible for hosting of master and slave clusters**
- **Can span multiple data centers and work across multiple regions**
- **Requires the Multi Data Centre Bridge component**

Tyk Deployment Models: On-Premise Architecture



Software Architectures

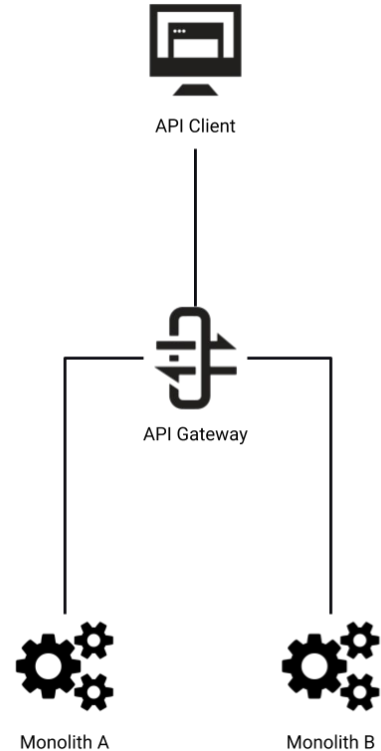
Software Architectures: Monolith

Monoliths are self-contained systems

**API Gateway enables communication
between consumers and monolithic
applications**

Transformation of protocols and data

Integration with legacy applications

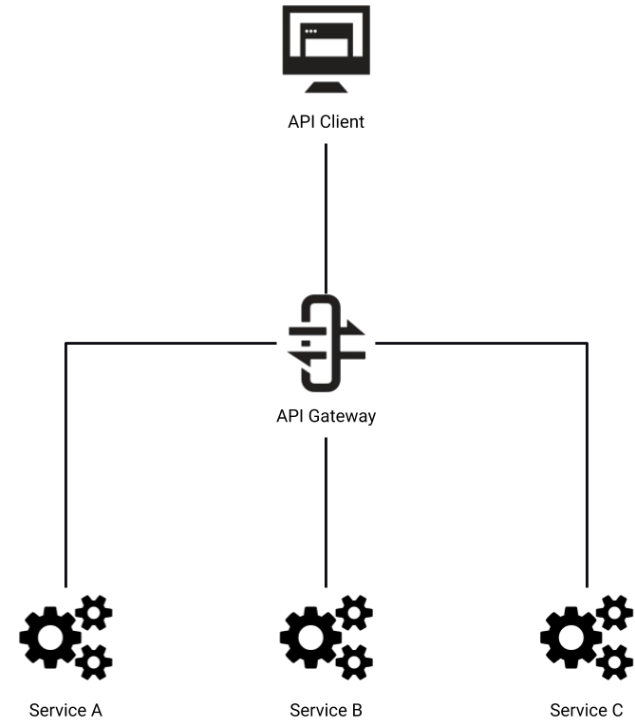


Software Architectures: Service Oriented Architecture

Services are discrete units of functionality

Centralised Gateway manages all traffic between consumers and services

Gateway implements common controls, such as authentication, authorisation, rate limiting and quotas



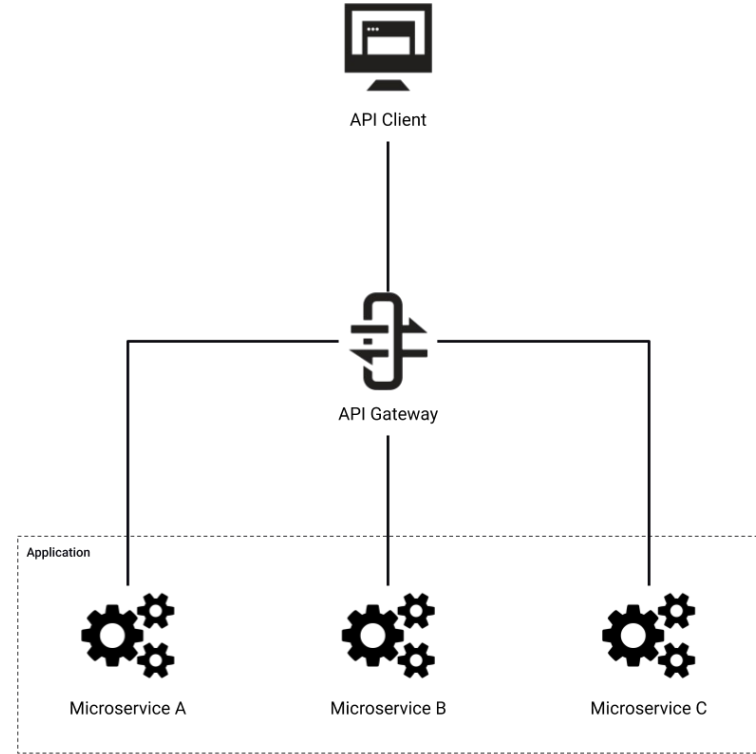
Software Architectures: Microservices

Architecturally similar to SOA, but services are decomposed to a smaller scope

Higher interconnectivity between systems

Benefits for Microservice architectures:

- **Distributed tracing**
- **Centralised logging**
- **Centralised analytics**
- **Access control**



Software Architectures: Service Mesh

Services are paired with Sidecar Gateways

- **Deployed within the same container**
- **Controlled by 3rd party tooling, such as Istio**

Ingress Gateway controlled by Tyk

Benefits for Service Mesh architectures:

- **Automation of traffic flows**
- **Centralised policy enforcement and monitoring**

